

COMICCRAFT

Phase 3 – Project Design

Project: ComicCraft – AI Comic Story Creator using Gemini Models

1. Purpose of the Design Phase

The Project Design Phase converts the requirements of ComicCraft into a structured application design. The source project describes three primary components: Frontend, Backend, and AI Integration. The frontend collects user inputs and displays comics, the FastAPI backend manages processing and routing, and the AI services generate story content and illustrations.

2. High-Level System Architecture

Layer	Main Responsibility	Technologies / Components
Frontend	Collect inputs and display generated comic	HTML, CSS, Jinja2
Backend	Routing, processing, workflow coordination	FastAPI, Python
AI Integration	Story and image generation	Gemini APIs, Stable Diffusion
Export	Compile comic into PDF	FPDF, generated images/text

3. Design Principle

The design separates user interaction, server-side processing, AI generation, and export responsibilities so that each major part of the comic-generation workflow can be handled as a distinct component.

Frontend Design

4. Frontend Architecture

The frontend is designed using HTML, CSS, and Jinja2 templates. The source document identifies three main templates: `index.html` for input collection, `comic_preview.html` for generated comic display, and `export_success.html` for export confirmation.

5. User Input Screen – `index.html`

- Story Prompt – main idea for the comic.
- Character Name – main character or hero.
- Setting – story location such as forest, school, city, or space.
- Story Tone – mood such as dramatic, funny, light-hearted, or poetic.
- Art Style – visual preference such as anime, pixel art, comic book, or realistic.

6. Comic Preview Screen – `comic_preview.html`

- Displays generated comic panels sequentially.
- Shows panel number and title.
- Displays the generated comic image.
- Shows scene description.
- Shows captions and narration.
- Displays the image prompt reference.

7. Export Success Screen – `export_success.html`

The design includes a separate success page that confirms the comic export/download and provides a path for the user to return and create another comic.

8. Template Data Flow

FastAPI routes use Jinja2 `TemplateResponse` rendering to dynamically bind generated comic data to the HTML templates. The input form maps its field names to backend parameters, while the preview template loops through the generated panel layout.

Backend & AI Integration Design

9. Backend Design

FastAPI is designed as the server-side layer responsible for receiving form or JSON input, coordinating AI functions, organizing the generated content, and supporting comic export.

10. Main Backend Workflow

Step	Component / Function	Design Role
1	User input	Receive story prompt and preferences.
2	generate_outline()	Create structured panel outline.
3	generate_story()	Create narration and dialogue.
4	generate_image()	Create panel illustrations.
5	build_comic_layout()	Match images with story content.
6	save_pdf()	Compile final comic into PDF.
7	Template response	Display result to user.

11. AI Integration Design

- Gemini Flash: structured comic outline generation.
- Gemini Pro: detailed narration and character dialogue.
- Stable Diffusion: comic-style illustration generation from image prompts.

12. AI Processing Relationship

The design connects text generation and image generation as a sequential workflow. The generated outline supplies the structure for the story, the story provides detailed panel content, and image prompts are used to generate corresponding illustrations. The backend then formats the resulting text and images for the frontend.

Routes, Layout & Export Design

13. Route Design

Route	Purpose
/	Load the homepage and user input form.
/generate	Process form input and generate comic preview.
/generate-comic/json	Accept JSON input and return comic layout/PDF information.
/test-image	Test image generation directly from a prompt.
/export-success	Display export completion confirmation.

14. Comic Layout Design

The layout builder is designed to associate each generated image with its corresponding panel story. The resulting layout contains panel information, image paths, and text. This structure is used for both preview rendering and export preparation.

15. PDF Export Design

The exporter compiles the comic into a multi-page PDF using FPDF. Each panel's image and narration are placed into the document, and the final PDF is saved in the export location specified by the project design.

16. End-to-End Design Flow

1	Input Form
2	FastAPI Route
3	Gemini Outline
4	Gemini Story
5	Stable Diffusion Images
6	Comic Layout
7	Jinja2 Preview
8	PDF Export

Project Structure & Design Summary

17. Proposed Component Structure

Based on the source documentation, ComicCraft is designed as a modular application where frontend templates, FastAPI routes, AI-generation modules, layout processing, and PDF export have distinct responsibilities.

Component	Design Responsibility
app/main.py	Application entry point and server configuration.
routes.py	FastAPI route handlers and request processing.
gemini_flash.py	Structured comic outline generation.
gemini_pro.py	Narration and dialogue generation.
image_generator.py	Comic illustration generation.
layout_builder.py	Panel/image/story layout organization.
exporters.py	PDF creation and export.
templates/index.html	User input interface.
templates/comic_preview.html	Generated comic preview.
templates/export_success.html	Export confirmation.

18. Design Outcome

The Project Design Phase defines how the ComicCraft requirements are translated into a working system. The design connects the user interface, FastAPI backend, Gemini text generation, Stable Diffusion image generation, layout processing, Jinja2 rendering, and PDF export into one continuous workflow.

19. Transition to Phase 4

The completed design provides the technical structure needed for Project Planning. Phase 4 can use this architecture to divide the work into milestones, activities, development tasks, testing tasks, and deployment preparation.

Source basis: ComicCraft – AI Comic Story Creator using Gemini Models project documentation.